

## Lab2: Insight into HTML and CSS

### Objective

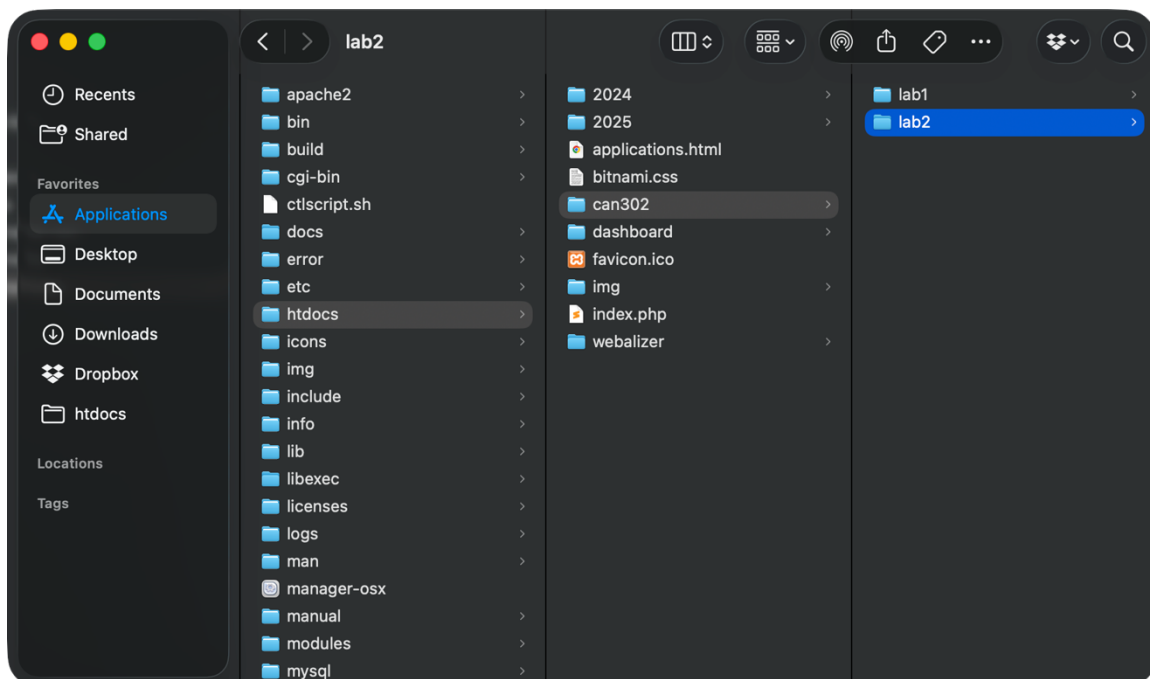
- Create and construct webpages using HTML
- Customize the style of webpages using CSS
- Describe the relationship between HTML and CSS

### Tips

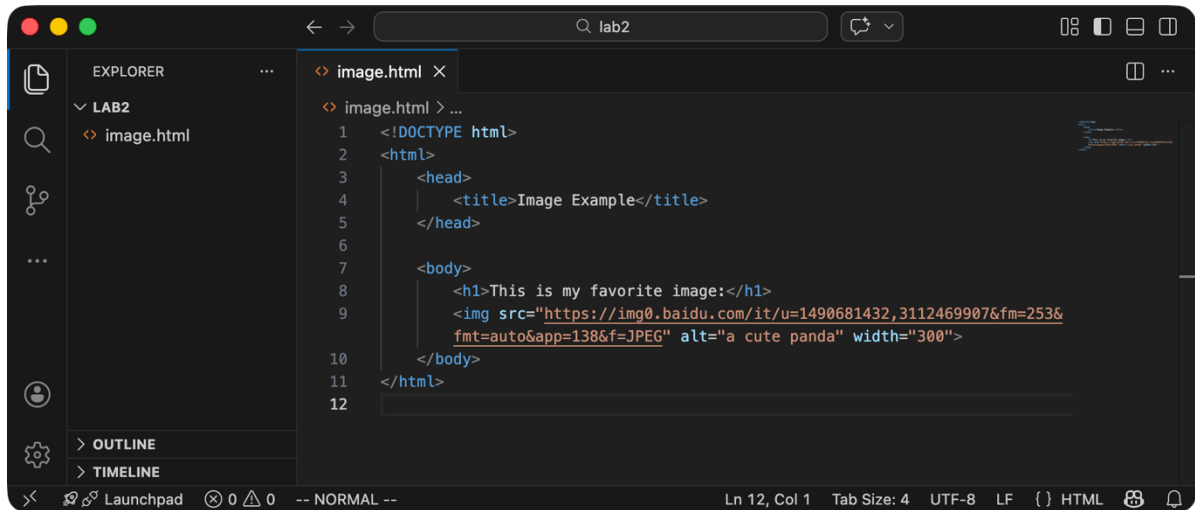
- These labs are expected to take more than the allocated 2 hours. You should try the tasks before the lab. Practice makes code perfect.
- Before the lab, you could post your questions in LMO forum, and you are welcome to offer solutions to the questions in LMO.
- During the lab, you could discuss with TAs for your questions. Face to face discussion will have a relatively high efficiency.

### A. HTML Basics

1. Create a new subfolder "lab2" under folder "can 302" (The following screenshot is in macOS, but same operations can be applied to Windows):



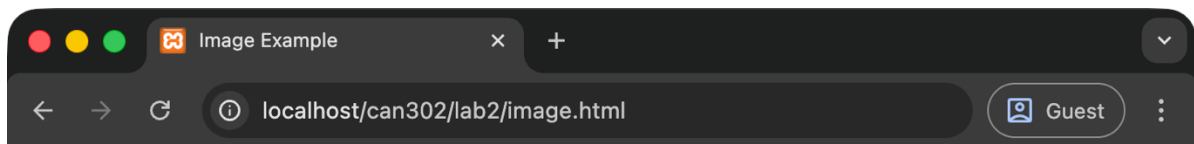
2. Use VS Code to open "lab2" and create a new file "image.html" as follows (the image link is: <https://img0.baidu.com/it/u=1490681432,3112469907&fm=253&fmt=auto&app=138&f=JPEG>):



The screenshot shows the Visual Studio Code editor with a file named "image.html" open. The Explorer sidebar on the left shows the file structure with "LAB2" and "image.html". The main editor area displays the following HTML code:

```
1 <!DOCTYPE html>
2 <html>
3   <head>
4     <title>Image Example</title>
5   </head>
6
7   <body>
8     <h1>This is my favorite image:</h1>
9     
11   </body>
12 </html>
```

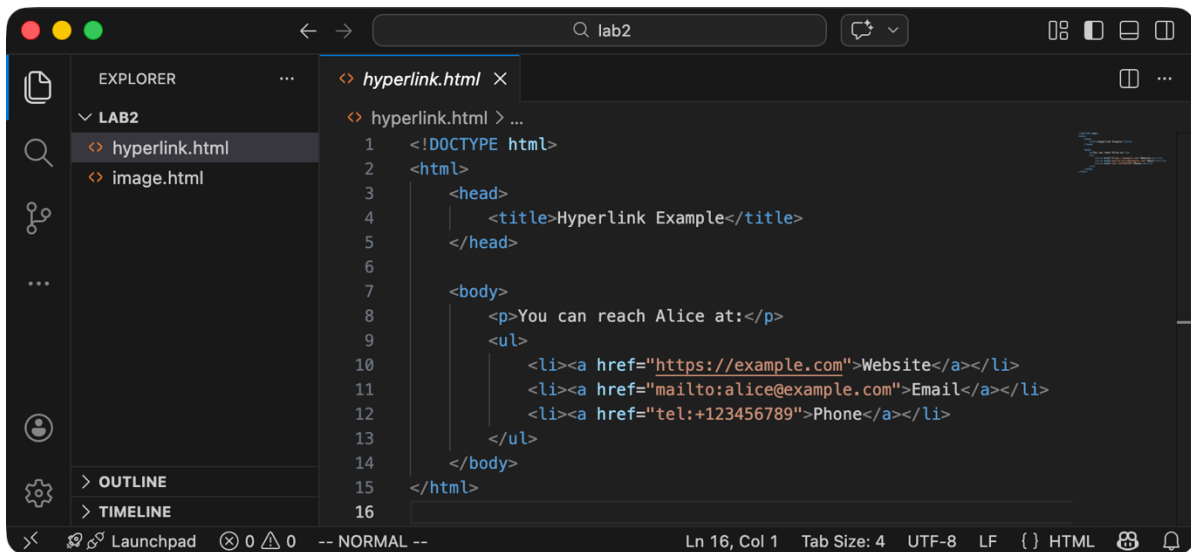
3. Access the webpage in your browser via URL: <http://localhost/can302/lab2/image.html>



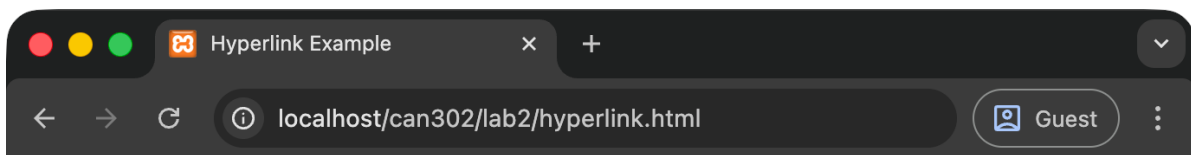
**This is my favorite image:**



4. Create a new file "hyperlink.html" under "lab2" as follows:



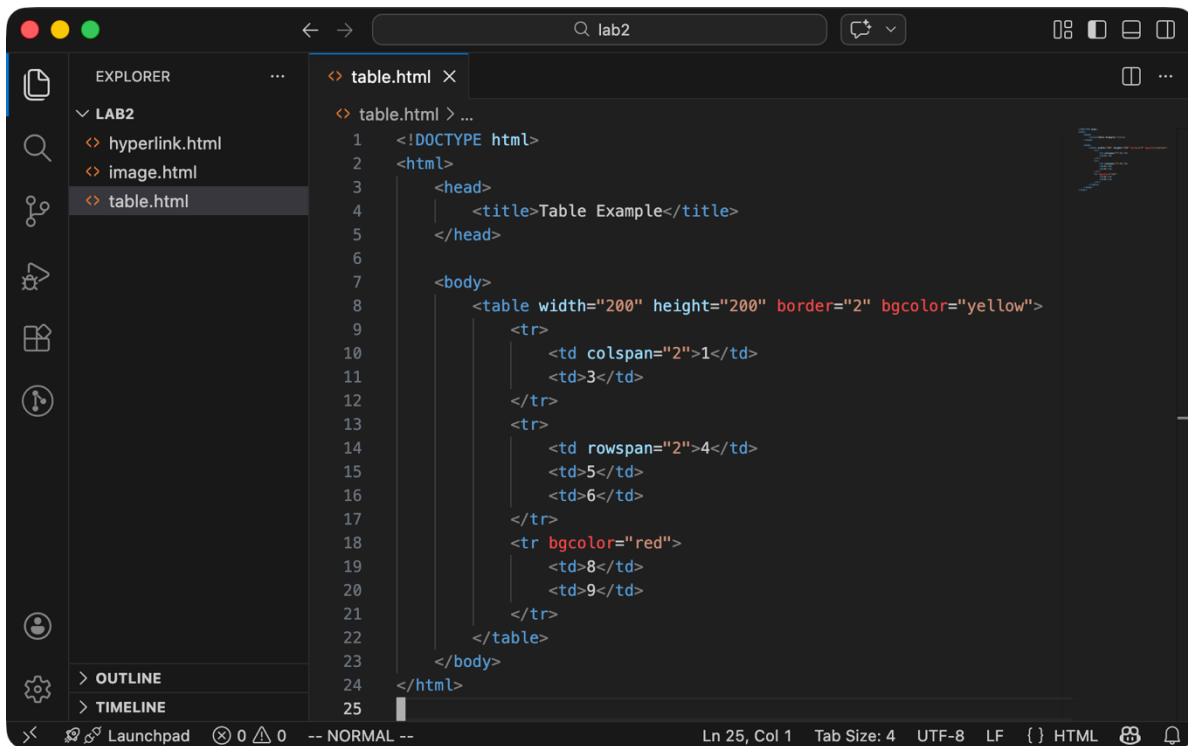
5. Access the webpage in your browser via URL: <http://localhost/can302/lab2/hyperlink.html>



You can reach Alice at:

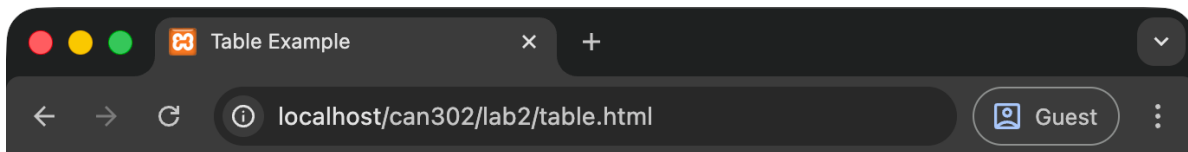
- [Website](https://example.com)
- [Email](mailto:alice@example.com)
- [Phone](tel:+123456789)

6. Create a new file "table.html" under "lab2" as follows:



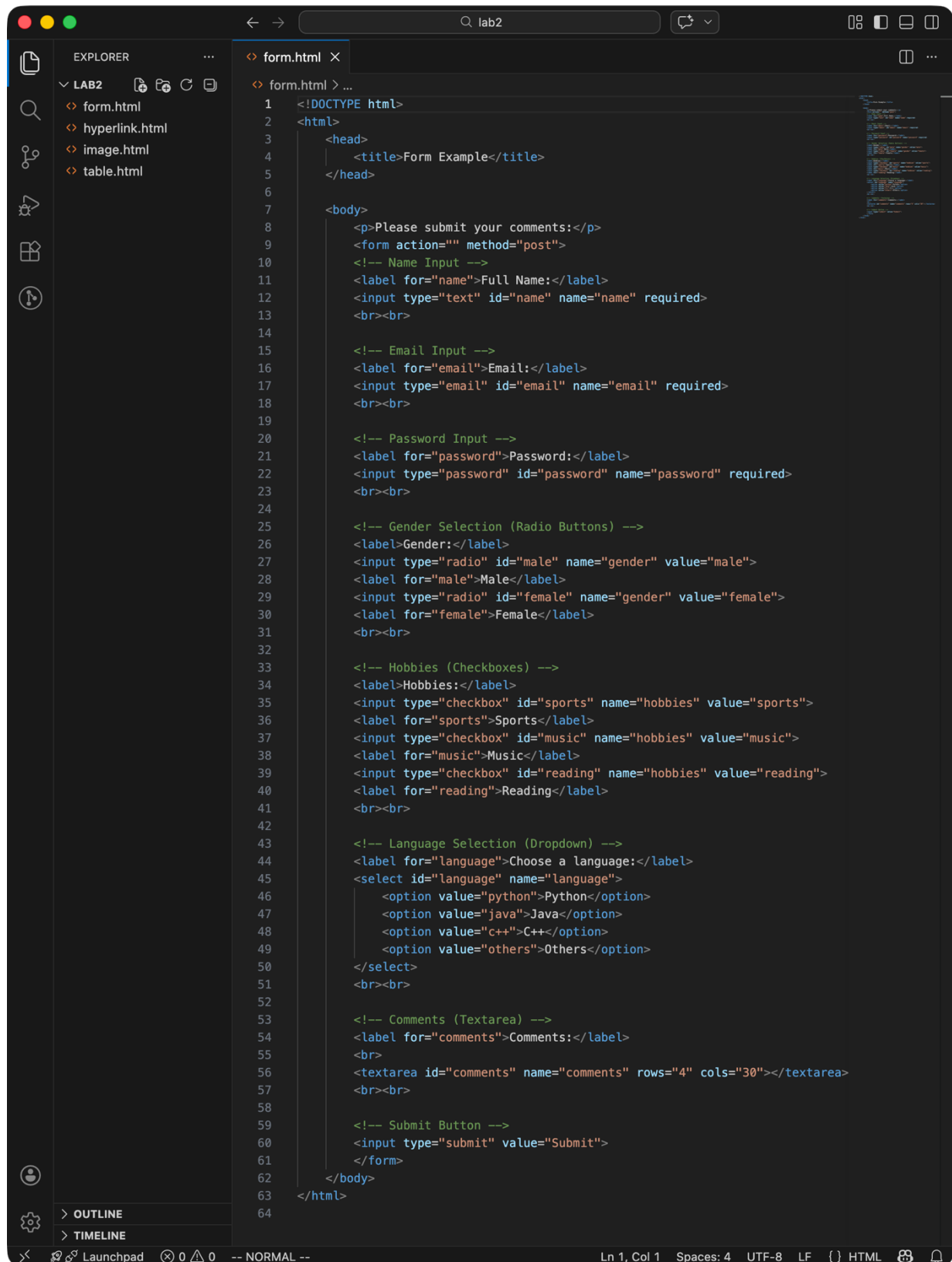
```
1 <!DOCTYPE html>
2 <html>
3   <head>
4     <title>Table Example</title>
5   </head>
6
7   <body>
8     <table width="200" height="200" border="2" bgcolor="yellow">
9
10      <tr>
11        <td colspan="2">1</td>
12        <td>3</td>
13      </tr>
14      <tr>
15        <td rowspan="2">4</td>
16        <td>5</td>
17        <td>6</td>
18      </tr>
19      <tr bgcolor="red">
20        <td>8</td>
21        <td>9</td>
22      </tr>
23    </table>
24  </body>
25 </html>
```

7. Access the webpage in your browser via URL: <http://localhost/can302/lab2/table.html>



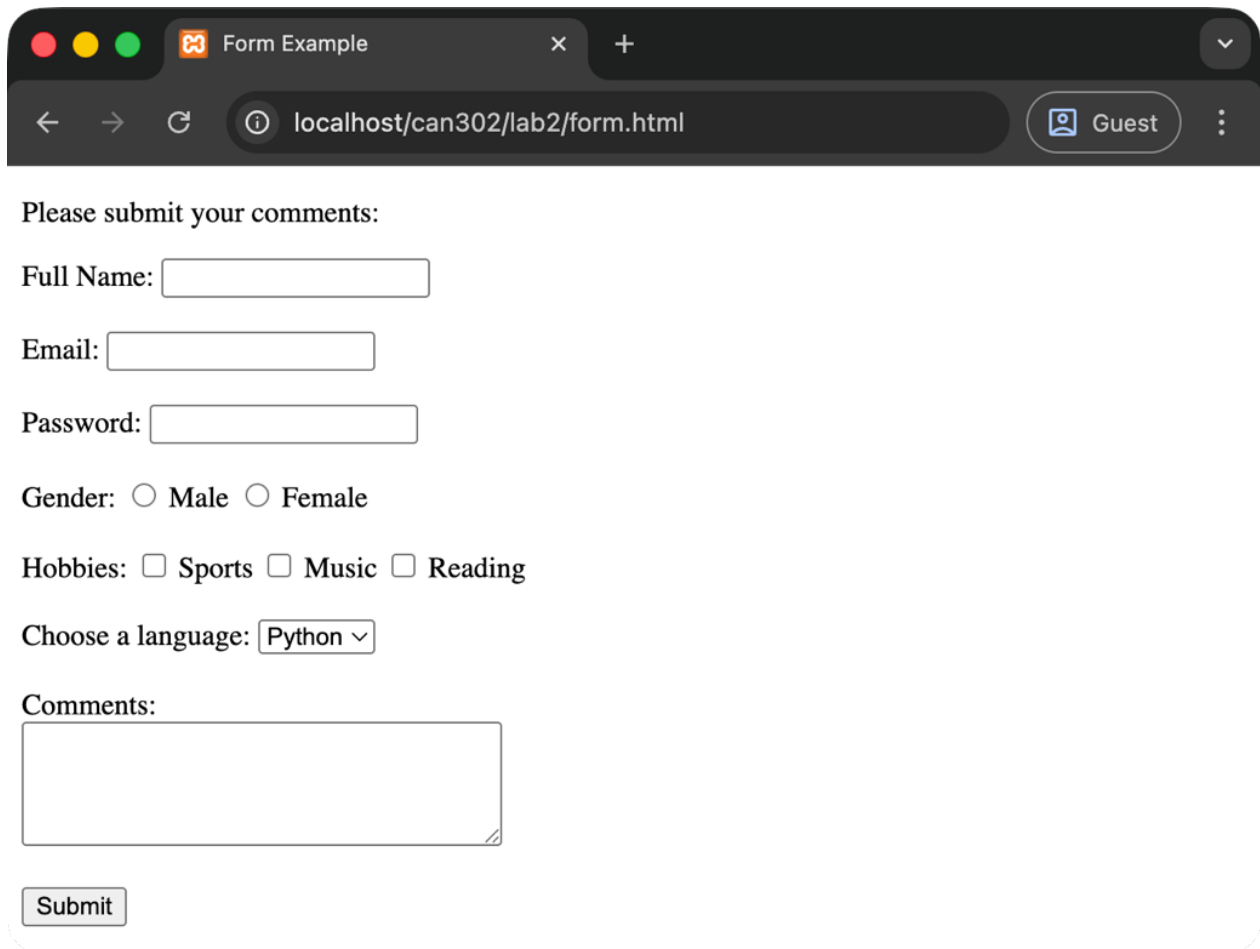
|   |   |   |
|---|---|---|
| 1 |   | 3 |
| 4 | 5 | 6 |
|   | 8 | 9 |

8. Create a new file "form.html" under "lab2" as follows:



```
1 <!DOCTYPE html>
2 <html>
3   <head>
4     <title>Form Example</title>
5   </head>
6
7   <body>
8     <p>Please submit your comments:</p>
9     <form action="" method="post">
10       <!-- Name Input -->
11       <label for="name">Full Name:</label>
12       <input type="text" id="name" name="name" required>
13       <br><br>
14
15       <!-- Email Input -->
16       <label for="email">Email:</label>
17       <input type="email" id="email" name="email" required>
18       <br><br>
19
20       <!-- Password Input -->
21       <label for="password">Password:</label>
22       <input type="password" id="password" name="password" required>
23       <br><br>
24
25       <!-- Gender Selection (Radio Buttons) -->
26       <label>Gender:</label>
27       <input type="radio" id="male" name="gender" value="male">
28       <label for="male">Male</label>
29       <input type="radio" id="female" name="gender" value="female">
30       <label for="female">Female</label>
31       <br><br>
32
33       <!-- Hobbies (Checkboxes) -->
34       <label>Hobbies:</label>
35       <input type="checkbox" id="sports" name="hobbies" value="sports">
36       <label for="sports">Sports</label>
37       <input type="checkbox" id="music" name="hobbies" value="music">
38       <label for="music">Music</label>
39       <input type="checkbox" id="reading" name="hobbies" value="reading">
40       <label for="reading">Reading</label>
41       <br><br>
42
43       <!-- Language Selection (Dropdown) -->
44       <label for="language">Choose a language:</label>
45       <select id="language" name="language">
46         <option value="python">Python</option>
47         <option value="java">Java</option>
48         <option value="c++">C++</option>
49         <option value="others">Others</option>
50       </select>
51       <br><br>
52
53       <!-- Comments (Textarea) -->
54       <label for="comments">Comments:</label>
55       <br>
56       <textarea id="comments" name="comments" rows="4" cols="30"></textarea>
57       <br><br>
58
59       <!-- Submit Button -->
60       <input type="submit" value="Submit">
61     </form>
62   </body>
63 </html>
64
```

9. Access the webpage in your browser via URL: <http://localhost/can302/lab2/form.html>



Please submit your comments:

Full Name:

Email:

Password:

Gender: ☐ Male ☐ Female

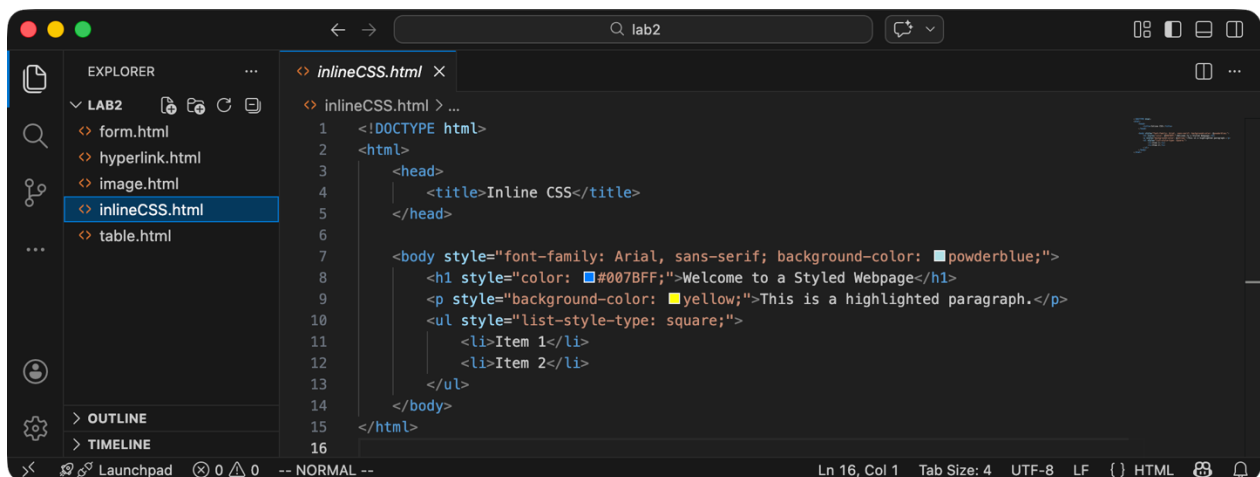
Hobbies: ☐ Sports ☐ Music ☐ Reading

Choose a language:

Comments:

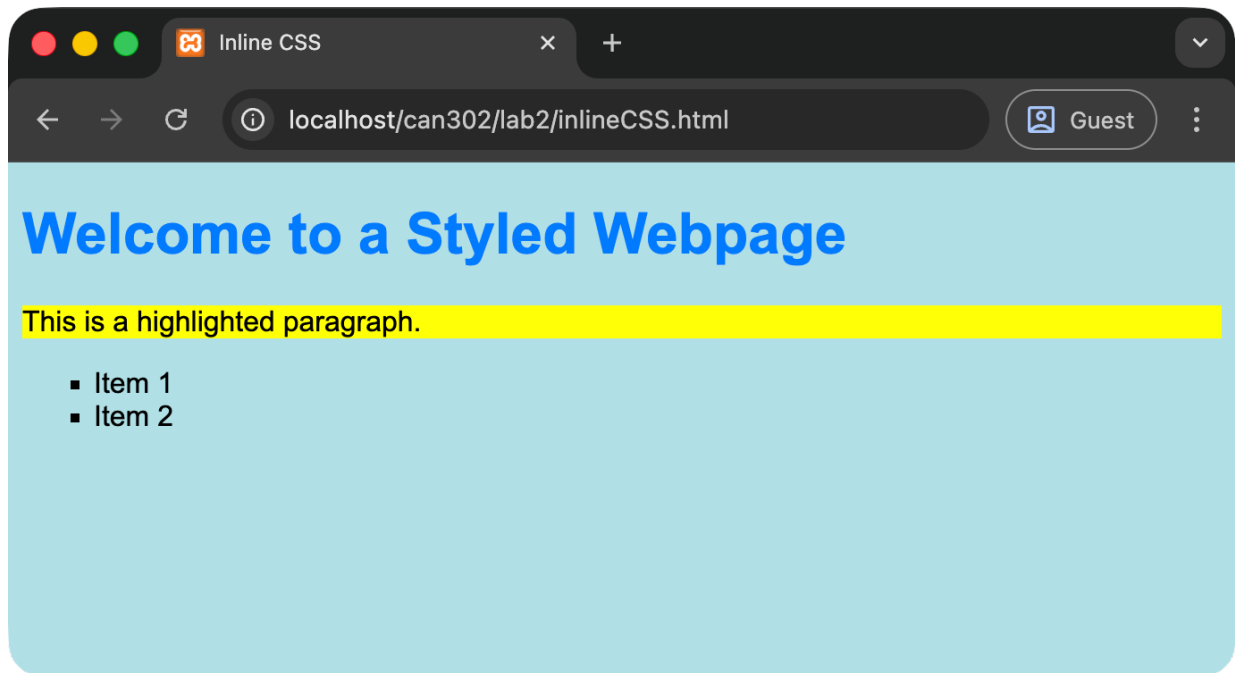
## B. CSS Basics

1. Create a new file "inlineCSS.html" under "lab2" as follows:

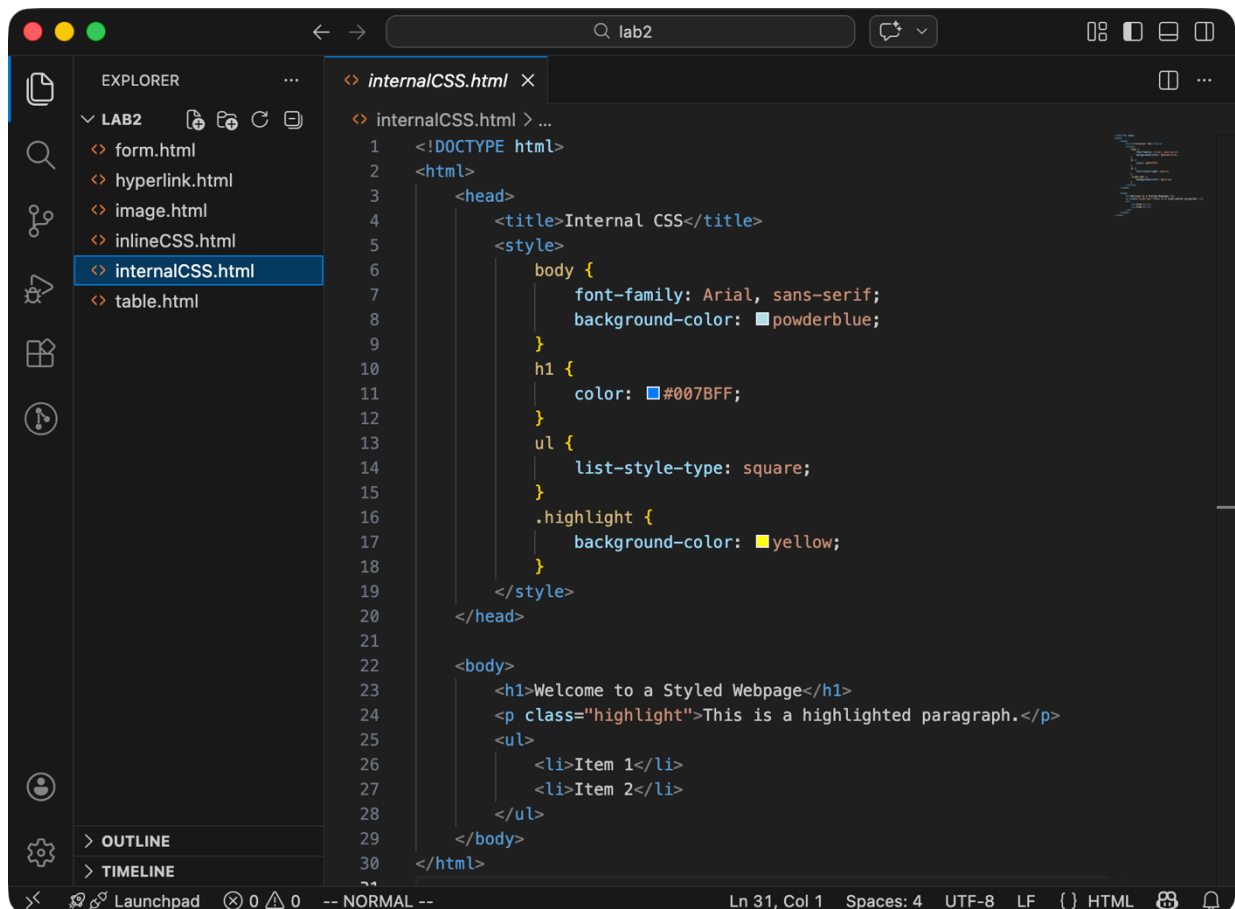


```
1 <!DOCTYPE html>
2 <html>
3   <head>
4     <title>Inline CSS</title>
5   </head>
6
7   <body style="font-family: Arial, sans-serif; background-color: powderblue;">
8     <h1 style="color: #007BFF;">Welcome to a Styled Webpage</h1>
9     <p style="background-color: yellow;">This is a highlighted paragraph.</p>
10    <ul style="list-style-type: square;">
11      <li>Item 1</li>
12      <li>Item 2</li>
13    </ul>
14  </body>
15 </html>
16
```

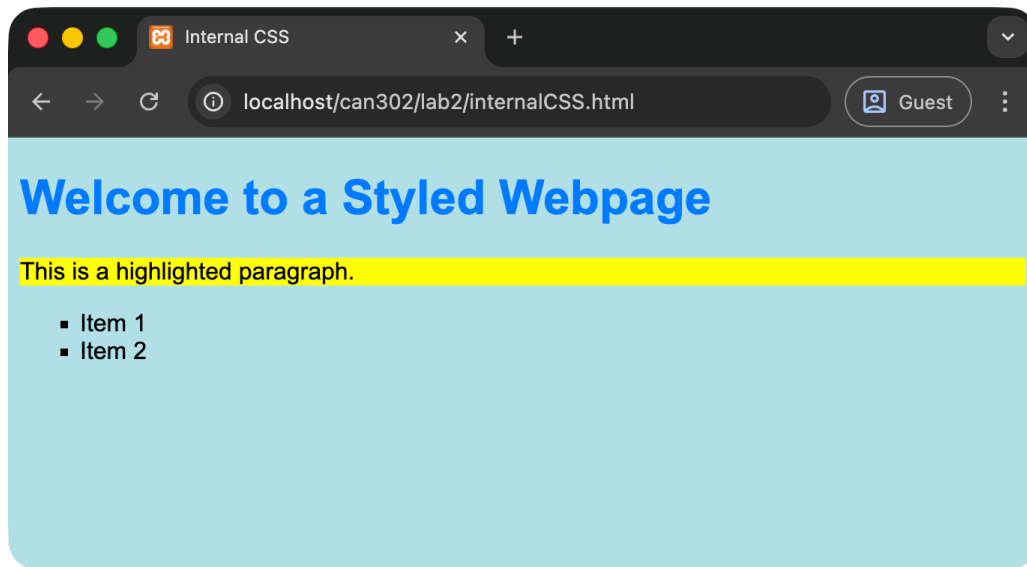
2. Access the webpage in your browser via URL: <http://localhost/can302/lab2/inlineCSS.html>



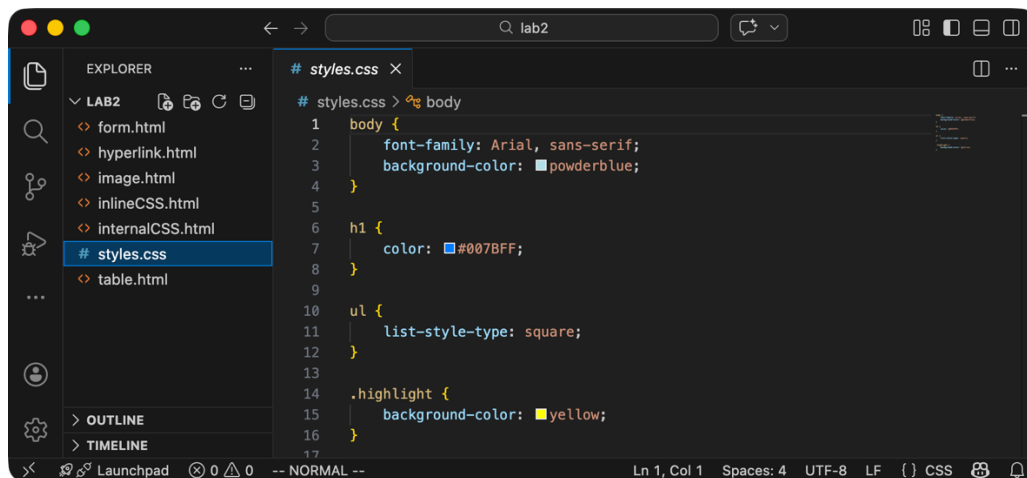
3. Create a new file "internalCSS.html" under "lab2" as follows:



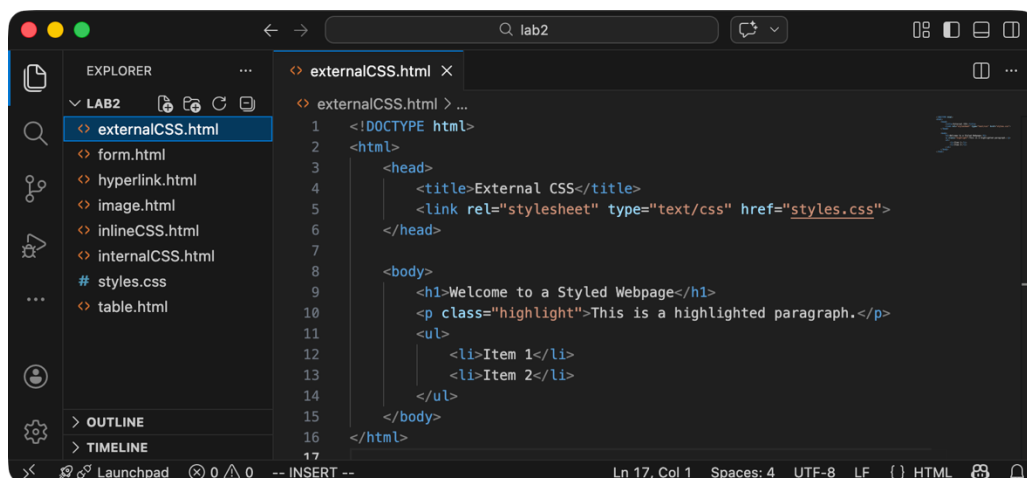
4. Access the webpage in your browser via URL: <http://localhost/can302/lab2/internalCSS.html>



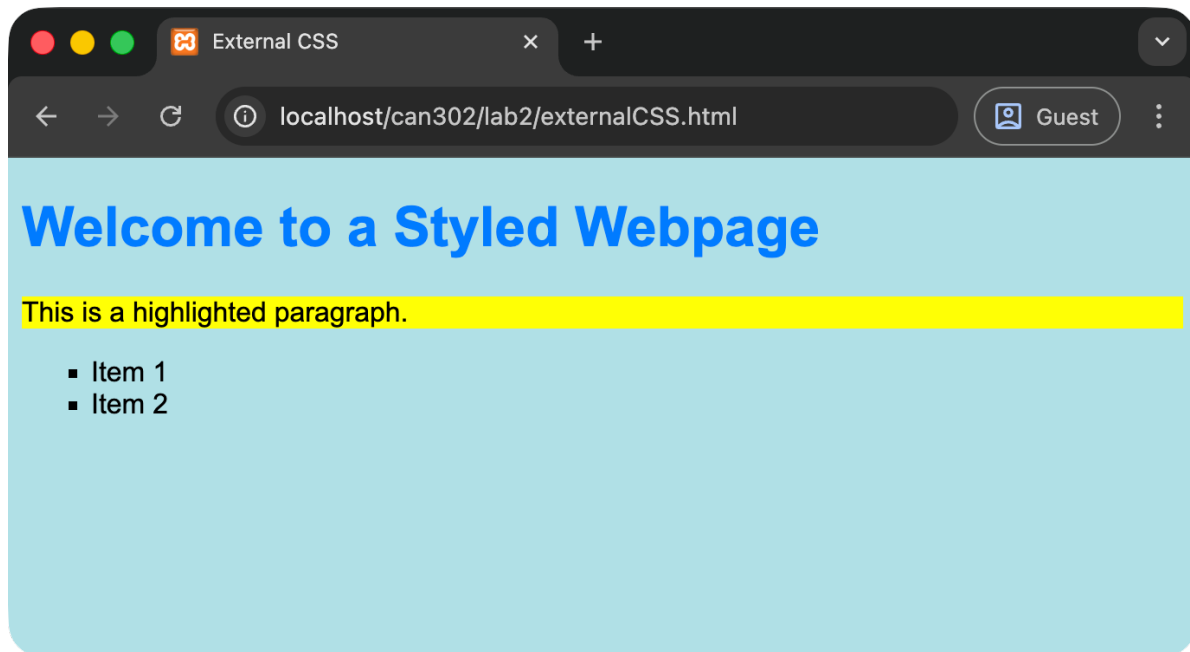
5. Create a new file "styles.css" under "lab2" as follows:



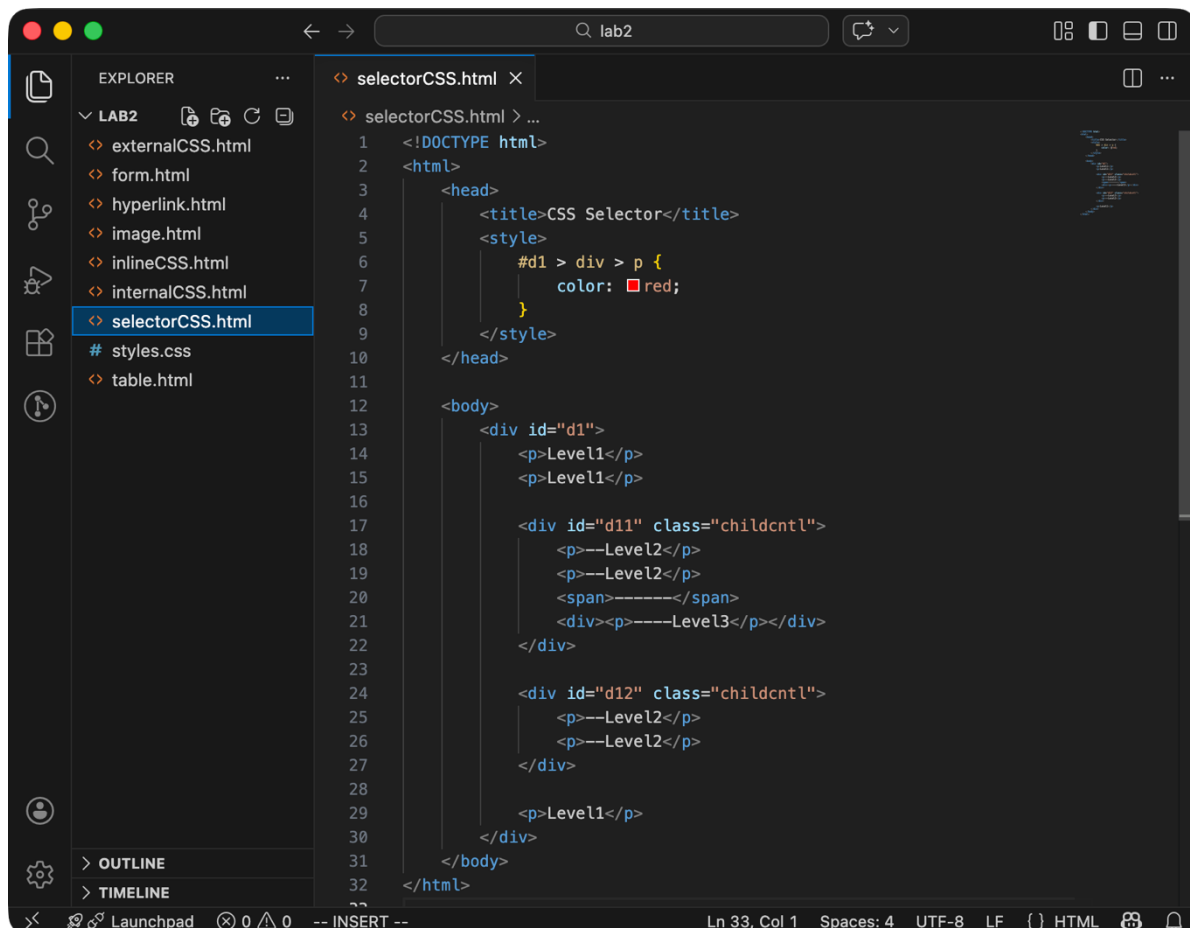
6. Create a new file "externalCSS.html" under "lab2" as follows:



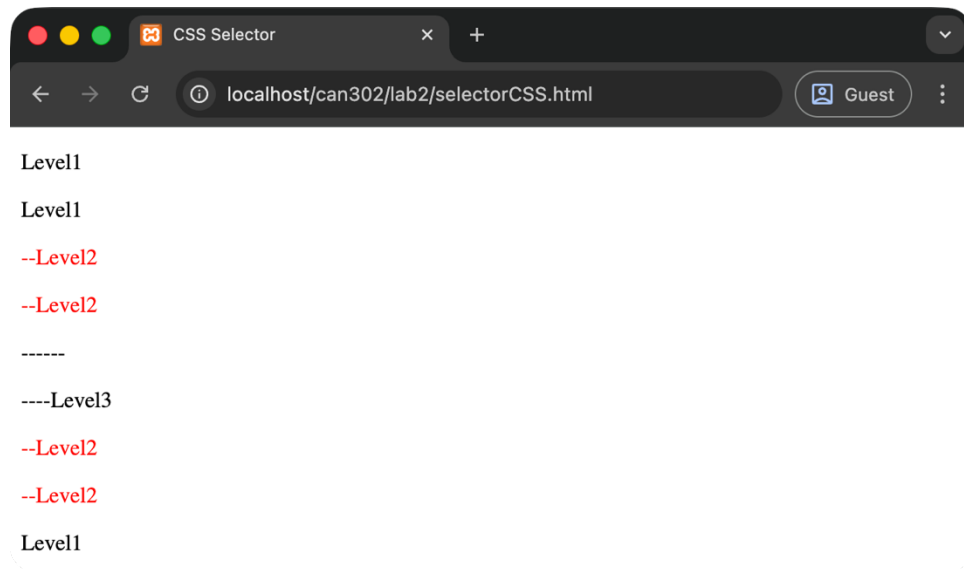
7. Access the webpage in your browser via URL: <http://localhost/can302/lab2/externalCSS.html>



8. Create a new file "selectorCSS.html" under "lab2" as follows:



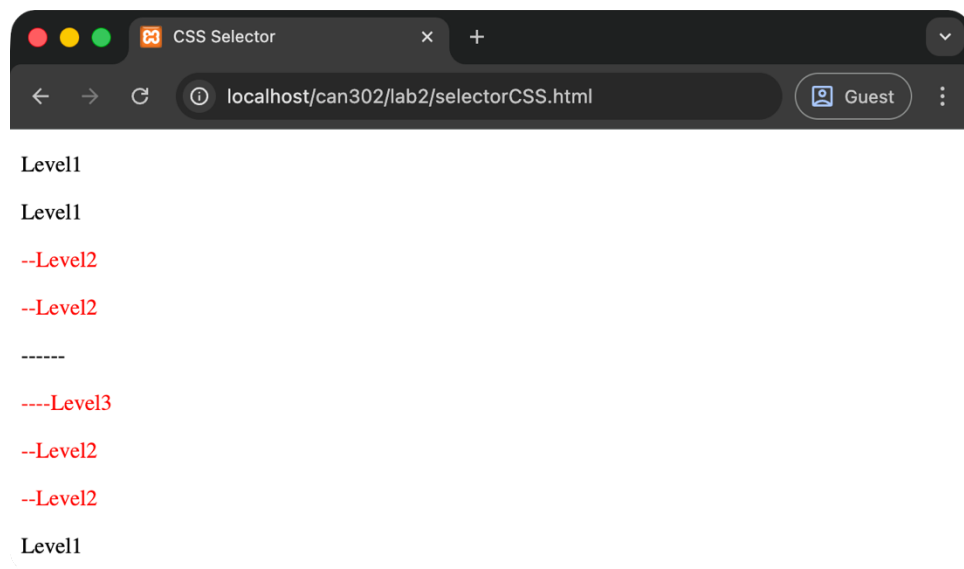
9. Access the webpage in your browser via URL: <http://localhost/can302/lab2/selecterCSS.html>



10. Change the lines 6—8 of "selectorCSS.html" as follows:

```
5 | <style>
6 |     #d1 div p {
7 |         color: red;
8 |     }
9 | </style>
```

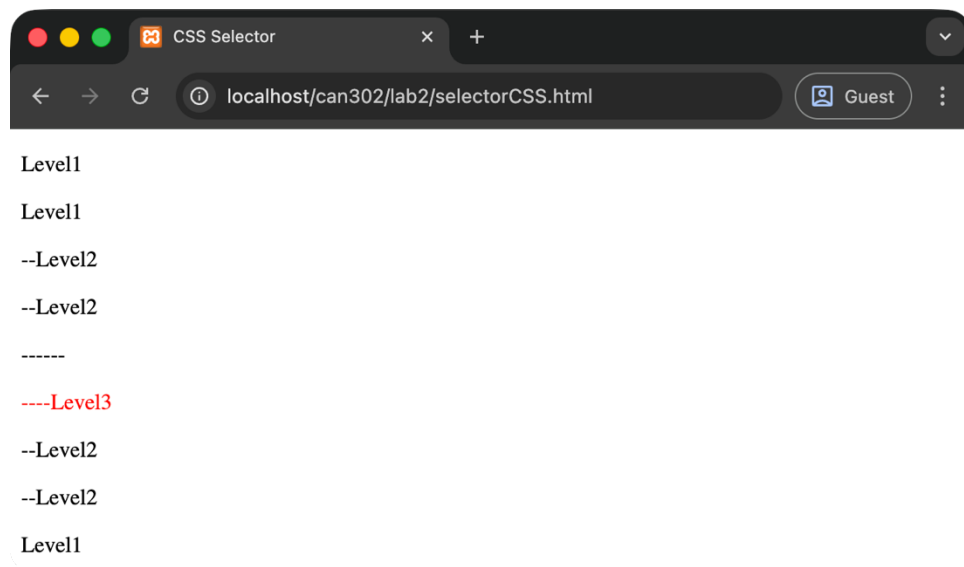
11. Reload the URL: <http://localhost/can302/lab2/selecterCSS.html>



12. Change the lines 6—8 of "selectorCSS.html" as follows:

```
5 | <style>
6 |     #d1 > div > div p {
7 |         color: red;
8 |     }
9 | </style>
```

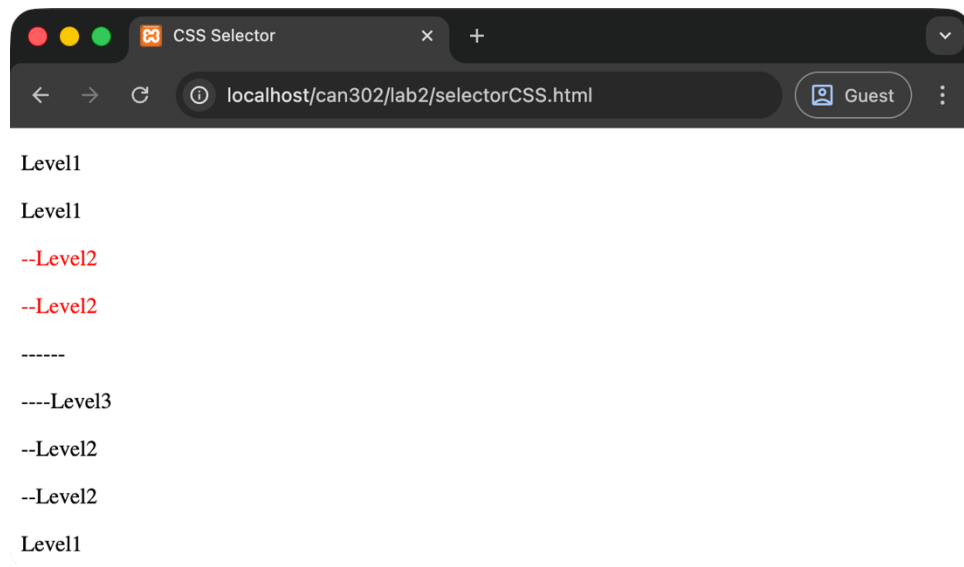
13. Reload the URL: <http://localhost/can302/lab2/selectoCSS.html>



14. Change the lines 6—8 of "selectorCSS.html" as follows:

```
5 | <style>  
6 |     p + div > p {  
7 |         color: red;  
8 |     }  
9 | </style>
```

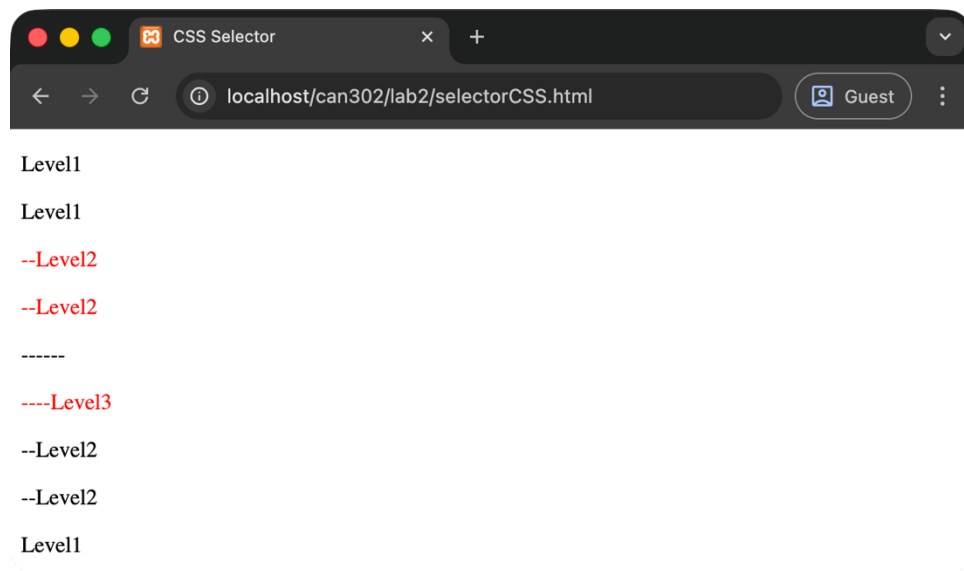
15. Reload the URL: <http://localhost/can302/lab2/selectoCSS.html>



16. Change the lines 6—8 of "selectorCSS.html" as follows:

```
5 | <style>  
6 |     p + div p {  
7 |         color: red;  
8 |     }  
9 | </style>
```

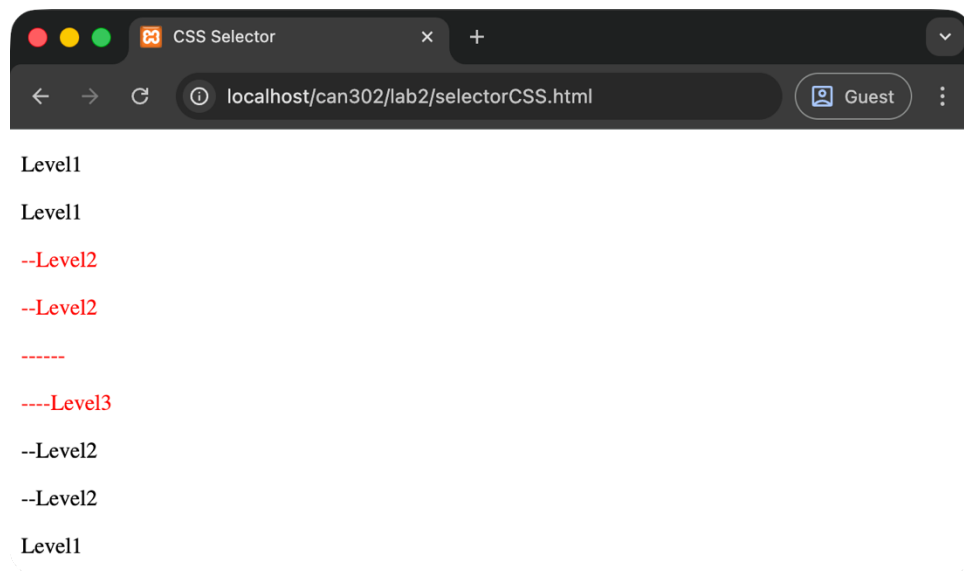
17. Reload the URL: <http://localhost/can302/lab2/selecterCSS.html>



18. Change the lines 6—8 of "selectorCSS.html" as follows:

```
5 | <style>
6 |     #d11.childcntl {
7 |         color: red;
8 |     }
9 | </style>
```

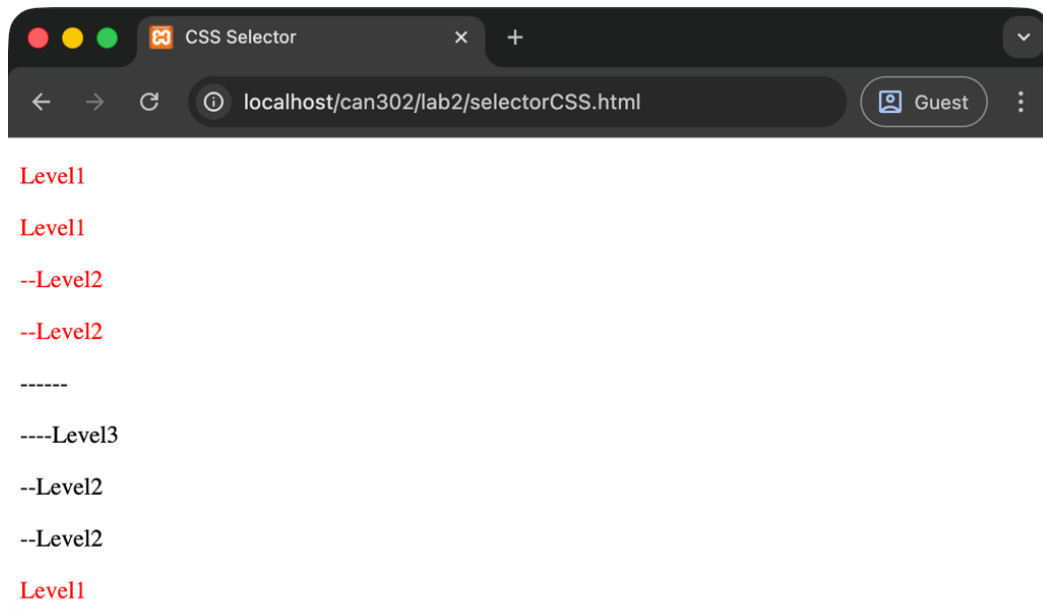
19. Reload the URL: <http://localhost/can302/lab2/selecterCSS.html>



20. Change the lines 6—8 of "selectorCSS.html" as follows:

```
5 | <style>
6 |     #d1 > p, #d11 > p {
7 |         color: red;
8 |     }
9 | </style>
```

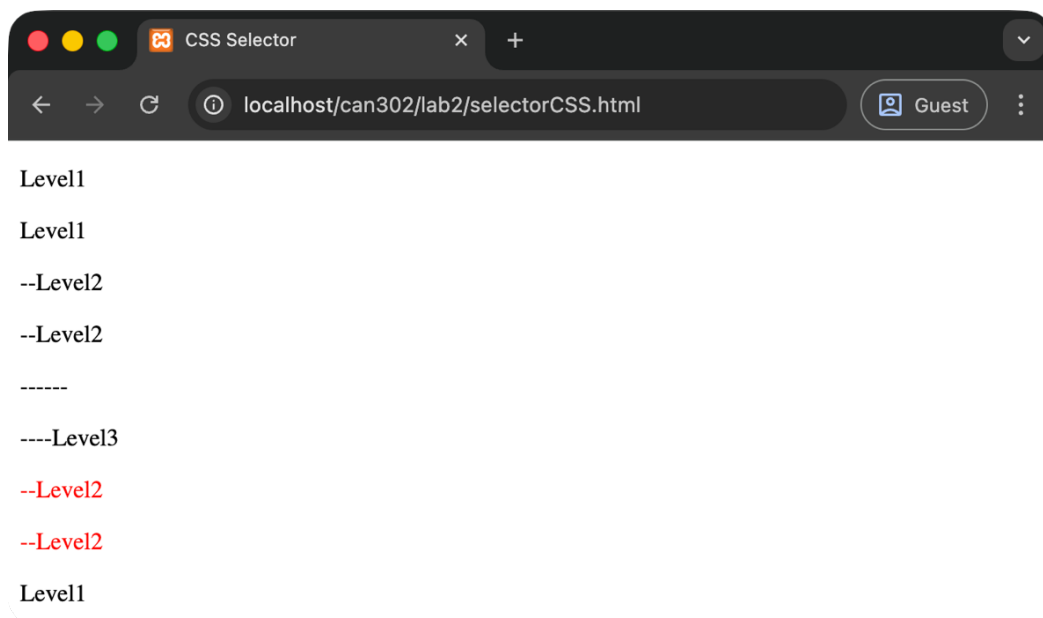
21. Reload the URL: <http://localhost/can302/lab2/selecterCSS.html>



22. Change the lines 6—8 of "selectorCSS.html" as follows:

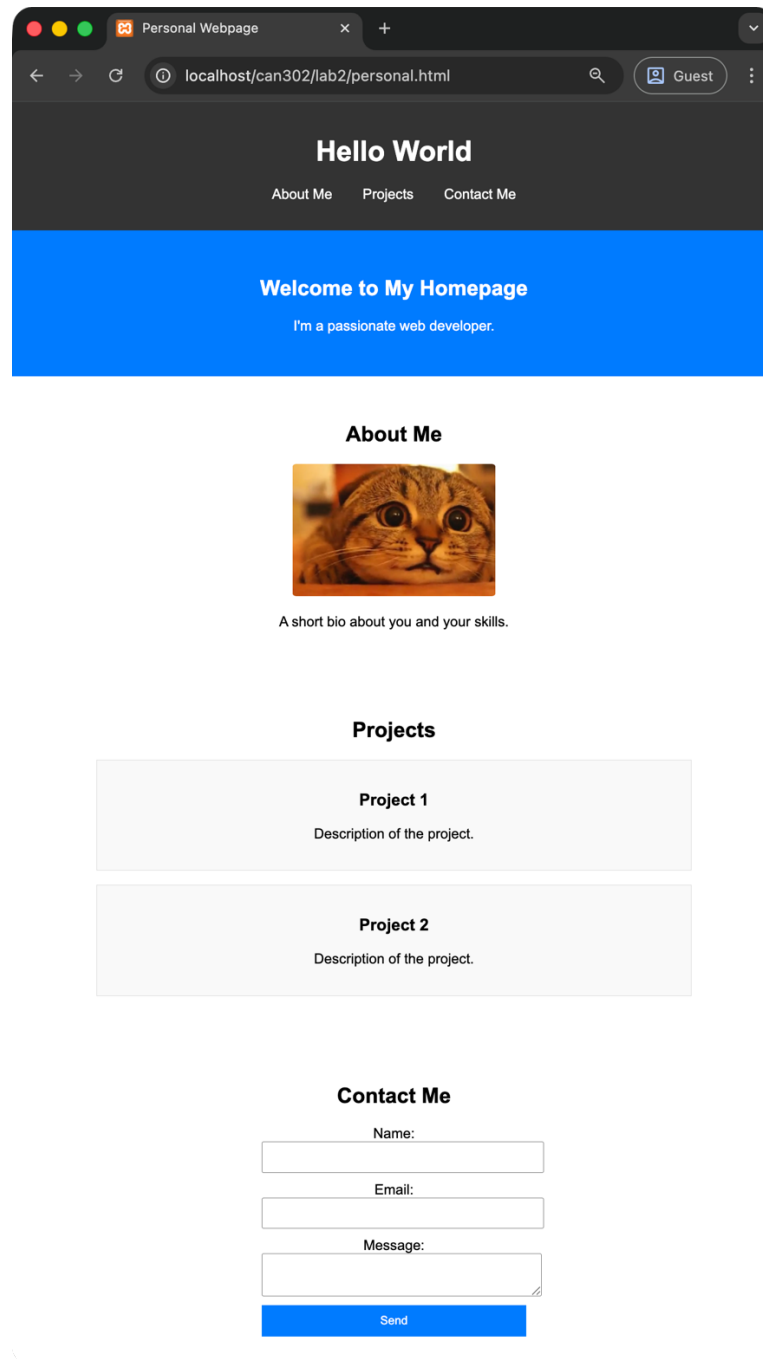
```
5 | <style>
6 |     #d11 + .childcntl {
7 |         color: red;
8 |     }
9 | </style>
```

23. Reload the URL: <http://localhost/can302/lab2/selecterCSS.html>



## C. Hand-on Exercises

1. Create an HTML page, which should contain the following information.
  - (a) Requirements:
    - Hyperlink to XJTLU homepage
    - Favorite image(s)
    - Favorite courses (at least 3)
    - Timetable of this semester
  - (b) Useful references:
    - [https://www.w3schools.com/tags/tag\\_a.asp](https://www.w3schools.com/tags/tag_a.asp)
    - [https://www.w3schools.com/tags/tag\\_img.asp](https://www.w3schools.com/tags/tag_img.asp)
    - [https://www.w3schools.com/html/html\\_lists.asp](https://www.w3schools.com/html/html_lists.asp)
    - [https://www.w3schools.com/html/html\\_tables.asp](https://www.w3schools.com/html/html_tables.asp)
2. Create a personal website with a responsive layout using HTML and CSS.
  - (a) Create a new file "personal.html" under "lab2" with the following sections:
    - A **header** with your name and a navigation menu
    - A **welcome section** with a short introduction and a profile picture
    - An **about section** with a brief bio
    - A **project section** with at least two projects (each having a title and description)
    - A **contact section** with a form (name, email, message)
  - (b) Create a new file "newstyles.css" under "lab2" (CSS styling):
    - Apply a consistent color scheme and typography to enhance aesthetics
    - Add hover effects on buttons and navigation links
  - (c) Useful references:
    - [https://www.w3schools.com/tags/tag\\_header.asp](https://www.w3schools.com/tags/tag_header.asp)
    - [https://www.w3schools.com/tags/tag\\_section.asp](https://www.w3schools.com/tags/tag_section.asp)
    - [https://www.w3schools.com/css/css\\_navbar.asp](https://www.w3schools.com/css/css_navbar.asp)
    - [https://www.w3schools.com/html/html\\_forms.asp](https://www.w3schools.com/html/html_forms.asp)
  - (d) An example:



## Summary

We have achieved the following things in this lab session:

- Explore the basics of HTML
- Explore the basics of CSS
- Explain the relationship between HTML and CSS